

# Agentic-seL4: Provably Secure Capability-Based Sandboxing of AI Agents

Modern systems are undergoing a seismic shift in their underlying technologies. On one hand, years of formal verification effort have produced major systems such as the **seL4** microkernel, which offers provably correct isolation and is now deployed in sectors from automotives (NIO) to trusted hypervisors (DornerWorks) [24], with the seL4-powered Kry10 OS (KOS) further extending adoption. On the other hand, these same sectors are being transformed by the rapid rise of **agentic AI**, which is being used to support tasks ranging from optimisation and self-healing to automated decision making [65]. However, agentic AI is inherently untrustworthy and greatly enlarges a system’s attack surface. Even during normal operation, an agent may inadvertently leak sensitive data, misconfigure systems, escalate its own privileges, or introduce subtle vulnerabilities in response to benign instructions. Under attack, it can actively amplify an adversary’s capabilities by following malicious prompts, exfiltrating data across trust boundaries, chaining tools and services, and autonomously weakening defences faster and more persistently than a human operator.

**Sandboxing AI agents** is therefore essential: we must tightly constrain their permissions, interactions, and observable state, and enforce robust isolation so that inevitable failures or compromises cannot escalate into systemic breaches. Unfortunately, this is also known to be highly challenging. Notable failures include Claude Mythos Preview (which escaped its sandbox in several controlled tests [4]), Snowflake Cortex (where a malicious README on GitHub was used to bypass a human approval gate [62, 64]), and NVIDIA OpenClaw (which was induced to exfiltrate sensitive data via “allowlisted” tools [59]). The UK AI Security Institute (AISi) showed that frontier models can readily exploit misconfigurations such as exposed Docker sockets, excessive privileges, and unpatched runtimes [27, 28, 29, 40].

The two **main goals** of AGENTIC-SEL4 are to

- (i) *provide provably secure sandboxing for systems comprising AI agents by using the capability-enabled seL4 microkernel as the foundation for isolation, and*
- (ii) *use AI agents to support the new security theorems that this extension induces, underpinned by modernisation of the existing seL4 theorems and proofs.*

Achieving these goals requires a new capability layer on top of seL4 (WP1) and an *agentic reference monitor* in user space that works with the trusted kernel to manage and control AI agent behaviour. This deliberately shifts part of the trust boundary into userland, but with behaviours that are carefully designed and formally specified. We will demonstrate the approach with a proof-of-concept implementation (WP2), and derive information-flow security theorems extending the original seL4 results to the new trust boundaries and dynamic agent–kernel interactions (WP3). While this will be an intensive time-bound proof and development effort with a very focussed goal, it will also produce widely reusable infrastructure: the seL4 proofs will be both modernised via a monadic Rely-Guarantee logic tailored to dynamic information-flow policies (extending the seL4 monadic Hoare logic) and accelerated through AI-assisted proof development in Isabelle/HOL, advancing neural theorem proving as a tool for scalable and maintainable formalisation (WP4).

The work will be conducted by a highly experienced **team** whose expertise spans information flow security, program and security logics, Isabelle/HOL and AI-based proof automation, AI security and natural language processing. On seL4, the Melbourne PI **led the seL4 information flow security proof effort**. We will collaborate with *Proofcraft*, whose team have 25 years of experience in extending and maintaining the seL4 proof and codebase (see letter of support).

---

## Prior Work

---

**Sandboxing AI.** AI containment is now recognised as a central challenge for building safe and secure agents, and hence, a range of sandboxing approaches have emerged. At the lowest level, Linux provides primitives such as bubblewrap (used by the Claude Code build system) to enable creation of sandboxed processes with carefully selected resource access. Other approaches use (lightweight) virtual machines or containers, e.g. Docker and Microsoft Execution Containers, to provide stronger resource isolation and finer-grained control over resource usage and networking. At a higher abstraction level, Nvidia’s OpenShell and NemoClaw support sandboxed, “always-on” and multi-agent systems.

The weaknesses of these approaches are, however, well documented [40], with vulnerabilities arising from incomplete enforcement [3], misconfiguration, flawed policies, and buggy runtimes. Once such a weakness exists, modern frontier AI models often manage to escape containment even without involvement of an actively malicious adversary. Sandbox-EscapeBench [40] addresses this by offering realistic escape scenarios and misconfigurations and measuring whether an agent can break containment. While valuable as an empirical stress test, such benchmarks are inherently post hoc and cannot deliver the kind of machine-checked isolation guarantees provided by systems like seL4.

**Capabilities for Security.** Capabilities are widely used to support access control (e.g., within operating systems). In seL4, capabilities can only be created by the kernel or derived from existing capabilities, but they may be delegated, restricted, or revoked entirely, making them an ideal, formally analysable foundation for sandboxing. Although the underlying mechanism is the same for AI and non-AI code, capabilities play a different role for agentic AI, which is a stochastic, prompt-driven model rather than a fixed program [41]. Agents can autonomously choose tools, chain

them in unforeseen ways, and be reprogrammed by (malicious) prompt injection, so over-broad capability grants are particularly dangerous. Capabilities therefore need to be finer-grained, dynamically delegated and revoked, and robust against active probing by the agent itself.

Several research systems already use capability-style mechanisms to defend against prompt injection. CaMeL (DeepMind and ETH) [15] attaches capability metadata to data and tools and uses a policy engine to check whether an agent’s side effects are permitted. Fides (Microsoft Research) [13] adds confidentiality and integrity labels, allowing tool calls only when label constraints are met. Nucleus [49] combines the lattice-based *portcullis* Rust capability library with an information-flow algebra to enforce end-to-end information-flow control, proving that untrusted content cannot silently reach privileged sinks, and providing a Lean formalisation with machine-checked proofs. However, none of these systems offer a seL4-style capability model at the OS level, so even if a runtime is secure up to its boundary, there are no end-to-end guarantees about the correctness of the underlying capability management or the surrounding system.

**Formalising and Verifying Information-Flow Security.** Security properties such as *noninterference* and *secure declassification* [34, 58] have been formalised in frameworks such as *hyperproperties* [12], *gradual release* [5] and our own *bounded-deducibility security* [51] plus knowledge-based formulations supported by deductive program logics [47]. Finer-grained relative/conditional policies have been developed, which prevent not all leakage, but only *additional* leakage through specific (notably microarchitectural) channels [11]; our recent *relative security* framework mechanized in Isabelle [16, 19] characterises leaks introduced by modified semantics (of which speculative execution is a particular case), and features techniques for verifying low-level binaries [25].

Security proofs for operating system kernels are increasingly common [14, 36, 71]. In earlier work, we showed that seL4 satisfies a variant of intransitive noninterference for the full verified kernel (almost 10k lines of C) [43], with seL4’s capability system as the key representation of authority. Prior results on authority confinement and integrity [60] established that the evolving capability state respects a prescribed access-control policy and that all kernel-mediated state changes are authorised by this policy. Building on this, we proved that the capability-based access-control structure induces a sound information-flow policy and obtained machine-checked confidentiality guarantees as intransitive noninterference. More recently, we generalised this framework to account for microarchitectural timing channels by enriching the system model with abstract shared hardware resources and timing-sensitive information-flow notions [61], although this has not yet been integrated into the seL4 verification stack.

There are numerous mechanised techniques for verifying information-flow security, and there have been sustained efforts towards scaling these techniques to practical systems [22, 35]. Our own work [16, 19, 25, 43, 61] used Isabelle/HOL to develop reusable frameworks for unwinding [20, 44, 54], binary analysis [26] and refinement-based security proofs [46]. **Isabelle/HOL as a Neural Theorem Prover.** Isabelle/HOL is an interactive theorem prover with a small trusted kernel, a mature proof language (Isar), and strong automation (e.g. *sledgehammer*) *Neural theorem proving* (NTP) for Isabelle builds on this foundation: AI-based components propose proofs [10], while Isabelle’s kernel and automation check correctness, so the neural part remains an assistant, not part of the trusted core.

Learning-enhanced proof automation for Isabelle/HOL has a long history, from MaLAREa SG1 [63] and ML-guided *sledgehammer* and MizAR [9, 31] (see [8] for a survey) to recent LLM-based systems. Frameworks such as Thor [30] and FVEL [37] combine enhance automation with LLMs, allowing independently trained models to assist new proofs. Baldur [23] uses Thor for whole-proof generation and repair, while Minilang [67] redesigns the proof language to a minimalist core on which LLMs significantly outperform Isar-trained baselines. Given the size and importance of the seL4 proof corpus (including C verification), several efforts have trained LLMs directly on this data, e.g. Selene [69], FVEL [37], and AutoReal [68]. These systems can already reprove substantial subsets of the theorems used in the seL4 correctness proofs, demonstrating both the feasibility and the promise of NTP for large industrial verification artefacts.

---

### Objectives and Work Packages

---

Our main goals will be achieved through the following four **objectives**, which give rise to corresponding **work packages**.

- O1.** Design an agent-aware runtime monitor for seL4 that supports dynamic creation, delegation, and revocation of capabilities, enabling fine-grained isolation and interaction with AI agents and their tools.
- O2.** Integrate and evaluate a runtime stack on top of seL4 that hosts realistic agentic AI, demonstrating practical sandboxing, capability management, and containment under adversarial and prompt-injection style scenarios.
- O3.** Verify new information-flow security theorems that account for agentic AI workloads, while preserving end-to-end noninterference and confinement guarantees. In conjunction, we will modernise the information flow security proofs to improve maintainability, including with the help of an LLM in the loop (O4).
- O4.** Advance Isabelle/HOL as an NTP platform by developing AI agents that can automatically maintain, extend, and refactor the seL4 proof base, and evaluate their impact on proof effort, robustness, and scalability.

**WP1: Agent-aware runtime monitor for seL4 (O1)** seL4’s current capability-based access control system is sufficient for providing kernel-level protections, even in the presence of agentic processes. However, to effectively support agentic workloads, we require an abstraction layer that serves as a runtime monitor for AI agents. The aim of this WP is to design this abstract layer and formalise its behaviour (cf. [42, 48]), while ensuring that it provides sufficient support for security policy reasoning. We anticipate the monitor may need to allow temporary privilege escalation or carefully managed access to sensitive data, moving some responsibility of capability management to userland processes.

**T1.1.** Analyse representative single-agent and multi-agent AI workflows to derive functional and security requirements for the runtime monitor, including support for dynamic tool acquisition, temporary authorities, and scoped delegation.

**T1.2.** Develop the API and rules for dynamically creating, delegating, and revoking capabilities, coupled with explicit contracts that are negotiated and enforced at runtime.

**T1.3.** Formalise the abstract specification of the runtime monitor, including proving it satisfies its top-level security specification that we intend to formalise in terms of capability-safety and dynamic information-flow security [17]: dynamic analogues of the existing seL4 security theorems that WP3 will generalise and extend.

**WP2: Agentic-seL4 runtime stack and evaluation (O2)** We will build and evaluate an experimental runtime stack for hosting realistic agentic AI workloads on top of seL4, demonstrating practical sandboxing and capability management. The runtime will map agent tools, data sources, and external services onto capability patterns, integrate representative agent frameworks, and include logging and monitoring for security evaluation. We will assess it using non-adversarial and prompt-injection-style scenarios to demonstrate end-to-end containment guarantees. We will develop a Linux-based system in parallel as a reference design for the seL4 implementation (see sprint structure below).

**T2.1.** Develop the runtime architecture, identifying the core components (agent process, tool wrappers, policy/contract manager, logging and monitoring), defining how each maps to seL4 protection domains and capabilities from WP1, and specifying the interaction patterns (IPC protocols, contracts for capability hand-off).

**T2.2.** Select one or more open-source agent frameworks (e.g. OpenAI Agents SDK, Open Interpreter OpenLLaMa) and encapsulate their tool interfaces in seL4-aware wrappers, replace ad hoc access control with capability-mediated calls, and add monitoring hooks for capability use and policy decisions.

**T2.3.** Work with the **Red Team**, to develop adversarial workloads based on SandboxEscapeBench, including *benign tasks* (e.g. code refactoring, document summarisation) to validate normal operation, and *adversarial tasks* (e.g., prompt injection) to stress containment. Analyse how capability use evolves and use these results to refine the runtime monitor (WP1), the **agentic AI as a threat model** and the formal capability models (WP1, WP3).

**WP3: Information-flow security theorems for agentic seL4 (O3).** We will revisit seL4’s existing information-flow security theorems [43] to extend them to cover the soundness of the reference monitor (WP1) and mechanically verify the resulting properties in Isabelle/HOL, developing security definitions tailored to agentic systems. The current seL4 security proofs (40KLOC) support only static intransitive noninterference policies [44], enforced via strict, statically bounded capability propagation [60], which must be generalised to support dynamic policies needed by AGENTIC-SEL4.

**T3.1.** Extend seL4’s existing access-control policy language [60] and its derived information-flow policy language [44] to support dynamic agentic behaviour. New constructs will capture temporary delegation of authority, permissions activated only after explicit approval, communication conditioned on holding an unrevoked capability or contract token, and policies that tighten as an agent accesses increasingly sensitive data. For example, we may have policies such as (1) an agent must not use external communication channels after reading a confidential file; (2) an agent that has read sensitive data must first undergo *model-context sanitisation* by a trusted component before any exfiltration step; and (3) in a multi-agent setting, a data-collection agent *A* may stream records to an analysis agent *B* only if *B* deletes its local copy on completion and a monitor agent *C* ensures no extra components are added to the stream.

**T3.2.** Formulate the security theorems—integrity enforcement, authority confinement, and information-flow security—as extensions of the existing seL4 results, relative to user-defined policy languages [44, 60], but now for the combined kernel/reference-monitor system rather than the kernel alone. We will gather evidence for the correctness of these formulations by testing them on toy examples and highly abstracted system models.

**T3.3.** Prove the above security theorems by leveraging seL4’s existing verified integrity and information-flow guarantees and relating them to the concrete runtime mechanisms from WP2. Our modular approach will first establish separate results for the kernel and the reference monitor, then combine them into end-to-end theorems using a general information-flow compositionality result (cf [6, 39] for exemplars in prior works).

**T3.4.** Iteratively **modernise** the existing seL4 proofs and work towards abstract frameworks that are more general and flexible, while being easier to maintain. (This is an overarching thread, taking place both before and during performing the main proofs.) In particular, we will: **(i)** Develop abstract relational models, yielding a *Monadic Relational Hoare*

*Logic with a rely-guarantee layer* that unifies integrity and information-flow proofs and verifies the reference monitor as a correct mediator of multi-agent interaction contracts. **(ii)** Instantiate our *bounded-deducibility security framework* [50, 51] for seL4 as a semantic basis for the dynamic information-flow policy language, using declassification bounds and triggers to capture behaviours such as model-context sanitisation by large untrusted components [2, 33]. **(iii)** Deploy the *relative security framework* [19, 54] to lift security properties from a base system to “obviously secure” extensions, allowing new reference-monitor functionality to be incorporated without redoing the entire verification effort.

**T3.5.** Instantiate the formal models with concrete WP2 scenarios to show how the theorems apply to realistic agentic configurations and how changes in capability/contract policies affect security guarantees. Document the structure, size, and maintainability of the resulting proof artefacts as input to WP4.

**WP4: Advancing Isabelle/HOL as an NTP platform (O4).** Building on the Isabelle AI infrastructure previously developed by the investigators—including training-data pipelines and the editor-agnostic PIDE MCP framework for interactive AI assistance in Isabelle [1]—we will adapt open-source LLMs (e.g., [38, 66]) through supervised fine-tuning on the seL4 proof base, followed by reinforcement learning using Isabelle’s verification feedback as reward. We will integrate these agents into a verification workflow so they can automatically maintain, extend, and refactor proofs (e.g., adapting to changes in the API, capability rules or security definitions). We will evaluate the impact of the use of agents on proof effort, robustness, and scalability. The goal is not just to automate isolated proofs, but to understand how neural proof assistants can safely and effectively participate in long-lived industrial-scale verification.

**T4.1.** Curate high-quality training corpora from the seL4 proof base by extracting, normalising and annotating representative proof fragments (particularly security-related components) with metadata such as dependencies, proof patterns and proof-state information. Develop Isabelle-specific data-generation pipelines to support supervised fine-tuning, reinforcement learning and systematic evaluation.

**T4.2.** Develop Isabelle proof agents capable of proposing complete proofs or proof sketches for new lemmas, repairing proofs broken by changes in reference monitor’s behaviour or in the security specifications, and suggesting maintainability-improving refactorings (e.g. lemma extraction or generalisation). Integrate these agents into Isabelle as optional, user-in-the-loop assistants.

**T4.3.** Use the evolving verification tasks from WP3 as benchmarks to measure the reduction in human proof effort (e.g. number of manual steps, engineering time), robustness of neural proofs to evolving specifications (e.g. after API changes), the quality of refactorings proposed by Isabelle/AI (proof size, structure, reuse).

**T4.4.** Develop principles for AI-assisted Isabelle verification, including proof-development patterns that facilitate effective AI assistance in long-lived verification developments, guidelines for incorporating AI assistance without expanding Isabelle’s trusted computing base, and an analysis of technical failure modes such as brittle proofs, overfitting to benchmarks, and failures to adapt robustly to evolving specifications.

---

### Sprints, Red Team Interaction and Future Extensions

---

AGENTIC-SEL4 is ambitious: it aims to give the highest possible insurance, based on a fully detailed semantically-grounded formal specification, of system-level security in the presence of AI agents, while using AI to boost theorem-proving productivity. This involves modernising a large proof codebase and verifying complex dynamic information-flow security theorems (WP3), which forms a *project-long* strand. By contrast, WP1, WP2 and WP4 involve rapid development and experimentation and form our *sprint* strand. We expect continual interaction between these strands: the long-term strand will yield better abstractions amenable to automation, while NTP advances (WP4) will feed back into progress on the core security proofs (WP3). Within the sprint strand, we will focus on delivering feedback via WP2 that informs the specification of the agentic runtime in WP1. We will also report on the experimental NTP results (WP4) and interim progress with the runtime (WP1) as we build knowledge of the agentic threat model.

Note that WP2 itself comprises two parallel implementations. Sprint 1 will focus on agentic integration with a capability OS, providing initial requirements for the runtime monitor (WP1). Sprints 2-4 will implement the runtime monitor on Linux, building experience for a seL4 implementation. The **Red Team**’s input will be used to stress test effectiveness under normal operation (without prompt injection). We expect the seL4 implementation to be ready by sprint 5; sprints 5-7 will stress test it under the same conditions as the Linux version. Simultaneously, we will explore capability protections under prompt-injection scenarios for the Linux implementation.

**Follow-on work.** This work paves the way for the development of an agentic monitor for systems such as KOS. Details of many other follow-on results are provided in our *commercialisation hypothesis*.

---

### Applicant Team and Project Management

---

AGENTIC-SEL4 brings together a diverse team with expertise in formal methods, AI, and security, comprising the investigators, 7 senior researchers, and 2 research engineers. We have already identified candidates for *all* funded posts. The investigators will devote substantial time over the project: Popescu (0.4 FTE), Dongol, Murray, and Valentino (0.3 FTE each), Aletras and Kaliszyk (0.2 FTE each). Thus the researchers benefit from close, hands-on supervision and can rapidly ramp up in this high-intensity effort. This level of commitment also allows the investigators to remain active technical contributors throughout, developing proofs and working directly with the proof agents alongside the research team.

**Prof Brijesh Dongol, University of Surrey, PI, Project Lead** is co-Director of the NCSC-funded Research Institute on Verified Trustworthy Software Systems (VeTSS) and an expert in formal methods for concurrent and distributed systems, program logics, memory models and security. His research includes the development of practical [21], memory-model aware program logics and formal verification frameworks in Isabelle [7, 18], for which he received a distinguished artefact award at ESOP 2022. He is an editor of the journals Formal Aspects of Computing and Science of Computer Programming and has led several large EPSRC projects. He will coordinate the project, lead WP2 and co-lead WP3.

**Prof Andrei Popescu, University of Sheffield, PI** has devoted his research career to designing compositional mechanisms for information-flow security and (co)inductive datatypes, and implementing them in Isabelle. His work on Isabelle reasoning infrastructure has received three POPL Distinguished Paper Awards. In 2021 he delivered an ITP keynote on formally verified information-declassification policies for realistic systems [50]. He has contributed 27 entries (300KLOC) to Isabelle's Archive of Formal Proofs and co-developed and verified CoCon [32, 52, 53], a conference management system with verified confidentiality, deployed for TABLEAUX 2015 and ITP 2016. He will lead WP4 and co-lead WP3.

**Prof Toby Murray, University of Melbourne, PI** is the director of the Defence Science Institute (Australia). His research is in formal methods for security and capability-based access control, with a recent focus on AI for formal verification. At NICTA he led the development of the sel4 information-flow security proofs [43], building on previous work on integrity enforcement [60] and later extending the proof base to account for timing leaks [61]. His sel4-related work has received major recognition, including the 2021 Australian Museum Eureka Prize for Outstanding Science in Safeguarding Australia and the 2022 ACM Software System Award. He has served as co-chair for CSF 2024/25 and the formal methods track at CCS 2026, and co-edited a special JCS issue on information-flow security. He will lead WP1 and co-lead WP3.

**Prof Cezary Kaliszyk, University of Melbourne, co-I** is an expert on automated theorem proving, formalised mathematics, learning for reasoning and proof guidance. His funded projects include Deeper4AI and SMART (ERC starting grant) which are developing machine learning guidance for modern state-of-the-art automatic theorem provers.

**Dr Marco Valentino, University of Sheffield, co-I** is an expert on AI systems for learning and reasoning in complex domains (in particular, evaluating and controlling LLMs) [57, 70]. His work has won a Best Resource Paper Award at EMNLP 2025 and an Outstanding Paper Award at EMNLP 2024. He also has expertise in practical Isabelle deployments in neurosymbolic settings [55, 56]. He will focus on WP2 and WP4.

**Prof Nikos Aletras, University of Sheffield, co-I** leads the Natural Language Processing (NLP) group in Sheffield, one of the strongest NLP groups in the UK. He is an expert on LLMs and has made recent breakthroughs in neurosymbolic reasoning. He will advise on developing the LLMs to support WP2 and WP4.

**Dr Aryan Pasikhani, University of Sheffield, co-I** is an expert on AI security, offensive AI and adversarial machine learning. He is the founder of CybPass, a University of Sheffield spin-out company that provides security testing for AI systems. He will support WP2 and coordinate the "Purple Team" liaising with the Red Team.

**Dr Gerwin Klein, Proofcraft, Collaborator** is Chief Scientist at Proofcraft and led the team that completed the initial verification of sel4. He has continued to maintain and extend its functionality, which now includes support for 20 platforms (within 5 different architectures) He will advise on WP1-WP3.

**Synergies.** Proofcraft have successfully maintained the proofs linked to the evolving sel4 code base for over 15 years despite substantial feature growth. We expect AGENTIC-SEL4 (in particular WP3/WP4) to support the maintainability and evolution of the broader sel4 proof ecosystem.

Dongol, Popescu, and Murray have previously collaborated on EPSRC-funded projects on Isabelle infrastructure and formal verification, yielding scalable, compositional information-flow frameworks (e.g., [6, 19, 45]), which will underpin the agent-governance mechanisms in this project. Popescu currently leads the project *Copilots for Isabelle: Learning Logical Structure for a Better Proving Experience*, which delivers an agentic setup for Isabelle, benefiting our planned sel4 proofs. That project is complementary and mutually reinforces AGENTIC-SEL4, with one advancing Isabelle AI technology and the other demonstrating it in a security-critical context. Valentino's research focuses on neurosymbolic methods and LLMs for theorem proving, with Isabelle as a primary target. Together with Aletras's NLP expertise, Kaliszyk's work on AI-assisted automated theorem proving, and Pasikhani's expertise in AI security and penetration testing, this creates a highly complementary team for AI-assisted large-scale formal verification and validation.

## References

- [1] K. Kappelmann with support from the rest of the “Copilots for Isabelle’ Team. *PIDE MCP: An Editor-Agnostic Model Context Protocol Server for Isabelle*. <https://github.com/kappelmann/isabelle-pide-mcp>. Open-source software developed at the University of Sheffield. 2026.
- [2] J. Andronick, D. Greenaway, and K. Elphinstone. “Towards Proving Security in the Presence of Large Untrusted Components”. In: *SSV*. Ed. by R. Huuck, G. Klein, and B. Schlich. USENIX Association, 2010. URL: <https://www.usenix.org/conference/ssv10/towards-proving-security-presence-large-untrusted-components>.
- [3] Anthropic. *How we contain Claude across products*. Retrieved: 2026-30-06. 2026. URL: <https://www.anthropic.com/engineering/how-we-contain-claude>.
- [4] Anthropic. *System Card: Claude Mythos Preview*. 2026. URL: <https://www-cdn.anthropic.com/08ab9158070959f88f296514c21b7facce6f52bc.pdf>.
- [5] A. Askarov and A. Sabelfeld. “Gradual Release: Unifying Declassification, Encryption and Key Release Policies”. In: *S&P*. IEEE Computer Society, 2007, pp. 207–221. DOI: 10.1109/SP.2007.22.
- [6] T. Bauereiß, A. Pesenti Gritti, **A. Popescu**, and F. Raimondi. “CoSMedis: A Distributed Social Media Platform with Formally Verified Confidentiality Guarantees”. In: *S&P*. IEEE Computer Society, 2017, pp. 729–748. DOI: 10.1109/SP.2017.24.
- [7] E. Vafeiadi Bila and **B. Dongol**. “A verified durable transactional mutex lock for persistent x86-TSO”. In: *Formal Methods Syst. Des.* 64.1 (2024), pp. 237–282. DOI: 10.1007/S10703-024-00462-1.
- [8] L. Blaauwbroek, D. M. Cerna, T. Gauthier, J. Jakubuv, **C. Kaliszyk**, M. Suda, and J. Urban. “Learning Guided Automated Reasoning: A Brief Survey”. In: *Logics and Type Systems in Theory and Practice - Essays Dedicated to Herman Geuvers on The Occasion of His 60th Birthday*. Ed. by V. Capretta, R. Krebbers, and F. Wiedijk. Vol. 14560. LNCS. Springer, 2024, pp. 54–83. DOI: 10.1007/978-3-031-61716-4\_4.
- [9] J.C. Blanchette, D. Greenaway, **C. Kaliszyk**, D. Kühlwein, and J. Urban. “A Learning-Based Fact Selector for Isabelle/HOL”. In: *J. Autom. Reasoning* 57.3 (2016), pp. 219–244. DOI: 10.1007/s10817-016-9362-8.
- [10] D. Bryant, J.J. Huerta y Munive, **C. Kaliszyk**, and J. Urban. “Munkres’ General Topology Autoformalized in Isabelle/HOL”. In: *Logic, Computation and Proof: Essays Dedicated to Larry Paulson on the Occasion of His 70th Birthday*. to appear. 2026. URL: <https://doi.org/10.48550/arXiv.2501.12345>.
- [11] S. Cauligi, C. Disselkoen, D. Moghimi, G. Barthe, and D. Stefan. “SoK: Practical Foundations for Software Spectre Defenses”. In: *S&P*. IEEE, 2022, pp. 666–680. DOI: 10.1109/SP46214.2022.9833707.
- [12] M. R. Clarkson and F. B. Schneider. “Hyperproperties”. In: *J. Comput. Secur.* 18.6 (2010), pp. 1157–1210. DOI: 10.3233/JCS-2009-0393.
- [13] M. Costa, B. Köpf, A. Kolluri, A. Paverd, M. Russinovich, A. Salem, S. Tople, L. Wutschitz, and S. Zanella-Béguelin. *Securing AI Agents with Information-Flow Control*. 2025. URL: <https://arxiv.org/abs/2505.23643>.
- [14] M. Dam, R. Guanciale, N. Khakpour, H. Nemati, and O. Schwarz. “Formal verification of information flow security for a simple ARM-based separation kernel”. In: *CCS*. 2013, pp. 223–234.
- [15] E. Debenedetti, I. Shumailov, T. Fan, J. Hayes, N. Carlini, D. Fabian, C. Kern, C. Shi, A. Terzis, and F. Tramèr. *Defeating Prompt Injections by Design*. 2025. URL: <https://arxiv.org/abs/2503.18813>.
- [16] J. Derrick, **B. Dongol**, C. Edmonds, M. Griffin, **A. Popescu**, and J. Wright. “Relative Security: (Dis)Proving Resilience Against Semantic Optimization Vulnerabilities in Isabelle/HOL”. In: *J. Autom. Reason.* 69.4 (2025), p. 31. DOI: 10.1007/S10817-025-09744-7.
- [17] C. Dimoulas, S. Moore, A. Askarov, and S. Chong. “Declarative policies for capability control”. In: *CSF*. IEEE. 2014, pp. 3–17.
- [18] **B. Dongol**. “Reasoning over Relaxed Shared Memory Models: A Tutorial”. In: *FM*. Ed. by A. Sampaio and M. Stoelinga. Vol. 16557. LNCS. Springer, 2026, pp. 394–419. DOI: 10.1007/978-3-032-26220-2\_20.
- [19] **B. Dongol**, M. Griffin, **A. Popescu**, and J. Wright. “Relative Security: Formally Modeling and (Dis)Proving Resilience Against Semantic Optimization Vulnerabilities”. In: *CSF*. IEEE, 2024, pp. 403–418. DOI: 10.1109/CSF61375.2024.00027.

- [20] **B. Dongol**, M. Griffin, **A. Popescu**, and J. Wright. “Secret-Directed Unwinding”. In: *Arch. Formal Proofs* 2024 (2024). URL: [https://isa-afp.org/entries/Secret\\_Directed\\_Unwinding.html](https://isa-afp.org/entries/Secret_Directed_Unwinding.html).
- [21] S. Egorov, G. V. Chockler, **B. Dongol**, D. O’Keeffe, and S. Keshavarzi. “Mangosteen: Fast Transparent Durability for Linearizable Applications using NVM”. In: *USENIX ATC*. Ed. by S. Bagchi and Y. Zhang. USENIX Association, 2024, pp. 799–815. URL: <https://www.usenix.org/conference/atc24/presentation/egorov>.
- [22] F. Erata, S. Deng, F. Zaghloul, W. Xiong, O. Demir, and J. Szefer. “Survey of Approaches and Techniques for Security Verification of Computer Systems”. In: *ACM J. Emerg. Technol. Comput. Syst.* 19.1 (2023), 6:1–6:34. DOI: 10.1145/3564785.
- [23] E. First, M. N. Rabe, T. Ringer, and Y. Brun. “Baldur: Whole-Proof Generation and Repair with Large Language Models”. In: *ESEC/FSE*. ESEC/FSE 2023. San Francisco, CA, USA: Association for Computing Machinery, 2023, pp. 1229–1241. ISBN: 9798400703270. DOI: 10.1145/3611643.3616243.
- [24] seL4 Foundation. *seL4 roadmap*. Retrieved: 2026-18-06. URL: <https://sel4.systems/roadmap.html>.
- [25] M. Griffin. “A Binary Analysis Platform in Isabelle/HOL”. PhD thesis. University of Surrey, 2024, p. 226. DOI: 10.15126/thesis.901307.
- [26] M. Griffin, **B. Dongol**, and A. Raad. “IsaBIL: A Framework for Verifying (In)correctness of Binaries in Isabelle/HOL”. In: *ECOOP*. Ed. by J. Aldrich and A. Silva. Vol. 333. LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2025, 14:1–14:30. DOI: 10.4230/LIPICS.ECOOP.2025.14.
- [27] UK AI Security Institute. *Can AI agents escape their sandboxes? A benchmark for safely measuring container breakout capabilities*. Retrieved: 2026-18-06. 2026. URL: <https://www.aisi.gov.uk/blog/can-ai-agents-escape-their-sandboxes-a-benchmark-for-safely-measuring-container-breakout-capabilities>.
- [28] UK AI Security Institute. *Inspect AI: Framework for Large Language Model Evaluations*. May 2024. URL: [https://github.com/UKGovernmentBEIS/inspect\\_ai](https://github.com/UKGovernmentBEIS/inspect_ai).
- [29] UK AI Security Institute. *The Inspect Sandboxing Toolkit: Scalable and secure AI agent evaluations*. Retrieved: 2026-18-06. 2025. URL: <https://www.aisi.gov.uk/blog/the-inspect-sandboxing-toolkit-scalable-and-secure-ai-agent-evaluations>.
- [30] A. Jiang, K. Czechowski, M. Jamnik, P. Milos, S. Tworkowski, W. Li, and Y. T. Wu. “Thor: Wielding Hammers to Integrate Language Models and Automated Theorem Provers”. In: *NeurIPS*. 2022.
- [31] **C. Kaliszyk** and J. Urban. “Mizar 40 for Mizar 40”. In: *J. Autom. Reason.* 55.3 (2015), pp. 245–256. DOI: 10.1007/S10817-015-9330-8.
- [32] S. Kanav, P. Lammich, and **A. Popescu**. “A Conference Management System with Verified Document Confidentiality”. In: *CAV*. Ed. by A. Biere and R. Bloem. Vol. 8559. Lecture Notes in Computer Science. Springer, 2014, pp. 167–183. DOI: 10.1007/978-3-319-08867-9\_11.
- [33] G. Klein, J. Andronick, M. Fernandez, I. Kuz, **T. C. Murray**, and G. Heiser. “Formally verified software in the real world”. In: *Commun. ACM* 61.10 (2018), pp. 68–77. DOI: 10.1145/3230627.
- [34] E. Kozyri, S. Chong, and A. C. Myers. “Expressing Information Flow Properties”. In: *Found. Trends Priv. Secur.* 3.1 (2022), pp. 1–102. DOI: 10.1561/33000000008.
- [35] T. Kulik, **B. Dongol**, P. G. Larsen, H. Daniel Macedo, S. Schneider, P. W. V. Tran-Jørgensen, and J. Woodcock. “A Survey of Practical Formal Methods for Security”. In: *Formal Aspects Comput.* 34.1 (2022), pp. 1–39. DOI: 10.1145/3522582.
- [36] S.-W. Li, X. Li, R. Gu, J. Nieh, and J. Z. Hui. “A secure and formally verified Linux KVM hypervisor”. In: *SP*. IEEE. 2021, pp. 1782–1799.
- [37] X. Lin, Q. Cao, Y. Huang, H. Wang, J. Lu, Z. Liu, L. Song, and X. Liang. “FVEL: Interactive Formal Verification Environment with Large Language Models via Theorem Proving”. In: *NeurIPS*. Ed. by A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang. Vol. 37. Curran Associates, Inc., 2024, pp. 54932–54946. DOI: 10.52202/079017-1742.
- [38] Y. Lin, S. Tang, B. Lyu, J. Wu, H. Lin, K. Yang, J. Li, M. Xia, D. Chen, S. Arora, and C. Jin. *Goedel-Prover: A Frontier Model for Open-Source Automated Theorem Proving*. 2025. URL: <https://arxiv.org/abs/2502.07640>.



- [39] H. Mantel. “A Uniform Framework for the Formal Specification and Verification of Information Flow Security”. PhD thesis. Universität des Saarlandes, 2003. URL: <https://publikationen.sulb.uni-saarland.de/handle/20.500.11880/25771>.
- [40] R. Marchand, A. O Cathain, J. Wynne, P. M. Giavridis, S. Deverett, J. Wilkinson, J. Gwartz, and H. Coppock. *Quantifying Frontier LLM Capabilities for Container Sandbox Escape*. 2026. URL: <https://arxiv.org/abs/2603.02277>.
- [41] E. Meijer. “Guardians of the Agents”. In: *Commun. ACM* 69.1 (Dec. 2025), pp. 46–52. ISSN: 0001-0782. DOI: 10.1145/3777544.
- [42] **T. C. Murray**. “Analysing the security properties of object-capability patterns”. PhD thesis. University of Oxford, UK, 2010. URL: <http://ora.ox.ac.uk/objects/uuid:98b0b6b6-eee1-45d5-b32e-d98d1085c612>.
- [43] **T. C. Murray**, D. Matichuk, M. Brassil, P. Gammie, T. Bourke, S. Seefried, C. Lewis, X. Gao, and G. Klein. “seL4: From General Purpose to a Proof of Information Flow Enforcement”. In: *S&P*. IEEE Computer Society, 2013, pp. 415–429. DOI: 10.1109/SP.2013.35.
- [44] **T. C. Murray**, D. Matichuk, M. Brassil, P. Gammie, and G. Klein. “Noninterference for operating system kernels”. In: *International Conference on Certified Programs and Proofs*. Springer, 2012, pp. 126–142.
- [45] **T. C. Murray**, R. Sison, and K. Engelhardt. “COVERN: A Logic for Compositional Verification of Information Flow Control”. In: *EuroS&P*. IEEE, 2018, pp. 16–30. DOI: 10.1109/EUROSP.2018.00010.
- [46] **T. C. Murray**, R. Sison, E. Pierzchalski, and C. Rizkallah. “Compositional Security-Preserving Refinement for Concurrent Imperative Programs”. In: *Arch. Formal Proofs* 2016 (2016). URL: [https://www.isa-afp.org/entries/Dependent\\_SIFUM\\_Refinement.shtml](https://www.isa-afp.org/entries/Dependent_SIFUM_Refinement.shtml).
- [47] **T. C. Murray**, M. Tiwari, G. Ernst, and D. A. Naumann. “Assume but Verify: Deductive Verification of Leaked Information in Concurrent Applications”. In: *CCS*. 2023, pp. 1746–1760.
- [48] J. Noble, A. Potanin, **T. C. Murray**, and M S. Miller. “Abstract and Concrete Data Types vs Object Capabilities”. In: *Principled Software Development - Essays Dedicated to Arnd Poetzsch-Heffter on the Occasion of his 60th Birthday*. Ed. by P. Müller and I. Schaefer. Springer, 2018, pp. 221–240. DOI: 10.1007/978-3-319-98047-8\_14.
- [49] Nucleus. Retrieved: 2026-20-06. 2026. URL: <https://github.com/coproduct-opensource/nucleus>.
- [50] **A. Popescu**, T. Bauereiss, and P. Lammich. “Bounded-Deducibility Security (Invited Paper)”. In: *ITP*. Ed. by Liron Cohen and Cezary Kaliszyk. Vol. 193. LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2021, 3:1–3:20. DOI: 10.4230/LIPICS.ITP.2021.3.
- [51] **A. Popescu** and P. Lammich. “Bounded-Deducibility Security”. In: *Arch. Formal Proofs* 2014 (2014). URL: [https://www.isa-afp.org/entries/Bounded\\_Deducibility\\_Security.shtml](https://www.isa-afp.org/entries/Bounded_Deducibility_Security.shtml).
- [52] **A. Popescu**, P. Lammich, and T. Bauereiss. “CoCon: A Confidentiality-Verified Conference Management System”. In: *Arch. Formal Proofs* 2021 (2021). URL: <https://www.isa-afp.org/entries/CoCon.html>.
- [53] **A. Popescu**, P. Lammich, and P. Hou. “CoCon: A Conference Management System with Formally Verified Document Confidentiality”. In: *J. Autom. Reason.* 65.2 (2021), pp. 321–356. DOI: 10.1007/S10817-020-09566-9.
- [54] **A. Popescu** and J. Wright. “Relative Security”. In: *Arch. Formal Proofs* 2024 (2024). URL: [https://www.isa-afp.org/entries/Relative\\_Security.html](https://www.isa-afp.org/entries/Relative_Security.html).
- [55] X. Quan, **M. Valentino**, D. S. Carvalho, D. Dalal, and A. Freitas. “PEIRCE: Unifying Material and Formal Reasoning via LLM-Driven Neuro-Symbolic Refinement”. In: *ACL (Volume 3: System Demonstrations)*. Ed. by P. Mishra, S. Muresan, and T. Yu. Association for Computational Linguistics, 2025, pp. 11–21. DOI: 10.18653/V1/2025.ACL-DEMO.2.
- [56] X. Quan, **M. Valentino**, L. A. Dennis, and A. Freitas. “Faithful and Robust LLM-Driven Theorem Proving for NLI Explanations”. In: *ACL (Volume 1: Long Papers)*. Ed. by W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar. Association for Computational Linguistics, 2025, pp. 17734–17755. DOI: 10.18653/V1/2025.ACL-LONG.867.



- [57] X. Quan, **M. Valentino**, L. A. Dennis, and A. Freitas. “Verification and Refinement of Natural Language Explanations through LLM-Symbolic Theorem Proving”. In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Ed. by Y. Al-Onaizan, M. Bansal, and Y.-N. Chen. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 2933–2958. DOI: 10.18653/v1/2024.emnlp-main.172. URL: <https://aclanthology.org/2024.emnlp-main.172/>.
- [58] A. Sabelfeld and D. Sands. “Declassification: Dimensions and principles”. In: *J. Comput. Secur.* 17.5 (2009), pp. 517–548. DOI: 10.3233/JCS-2009-0352.
- [59] Lasso Security. *Thinking Outside The Box: Exfiltrating OpenClaw Data from NVIDIA’s new Sandbox*. Retrieved: 2026-18-06. 2026. URL: <https://www.lasso.security/blog/sandboxed-ai-agents-attack-surface>.
- [60] T. Sewell, S. Winwood, P. Gammie, **T. C. Murray**, J. Andronick, and G. Klein. “seL4 Enforces Integrity”. In: *ITP*. Ed. by M. C. J. D. van Eekelen, H. Geuvers, J. Schmaltz, and F. Wiedijk. Vol. 6898. LNCS. Springer, 2011, pp. 325–340. DOI: 10.1007/978-3-642-22863-6\_24.
- [61] R. Sison, S. Buckley, **T. Murray**, G. Klein, and G. Heiser. “Formalising the Prevention of Microarchitectural Timing Channels by Operating Systems”. In: *FM*. Ed. by M. Chechik, J.-P. Katoen, and M. Leucker. Vol. 14000. LNCS. Springer, 2023, pp. 103–121. DOI: 10.1007/978-3-031-27481-7\_8.
- [62] Snowflake. *PromptArmor Report - Snowflake Response*. Retrieved: 2026-18-06. 2026. URL: <https://community.snowflake.com/s/article/PromptArmor-Report---Snowflake-Response>.
- [63] J. Urban, G. Sutcliffe, P. Pudlák, and J. Vyskočil. “MaLAREa SG1 - Machine Learner for Automated Reasoning with Semantic Guidance”. In: *IJCAR*. IJCAR '08. Sydney, Australia: Springer-Verlag, 2008, pp. 441–456. ISBN: 9783540710691. DOI: 10.1007/978-3-540-71070-7\_37.
- [64] Common Vulnerabilities and Exposures. *CVE-2026-6442*. Retrieved: 2026-18-06. 2026. URL: <https://www.cve.org/CVERecord?id=CVE-2026-6442>.
- [65] G. Xenos and D. Serpanos. “Agentic AI: A New Paradigm for Cyber-Physical Systems Cybersecurity”. In: *Computer* 59.01 (Jan. 2026), pp. 180–183. ISSN: 1558-0814. DOI: 10.1109/MC.2025.3625738.
- [66] H. Xin, D. Guo, Z. Shao, Z. Ren, Q. Zhu, B. Liu, C. Ruan, W. Li, and X. Liang. *DeepSeek-Prover: Advancing Theorem Proving in LLMs through Large-Scale Synthetic Data*. 2024. arXiv: 2405.14333 [cs.AI]. URL: <https://arxiv.org/abs/2405.14333>.
- [67] Q. Xu, R. Wang, P. Wang, H. Li, and C. Watt. “A Minimalist Proof Language for Neural Theorem Proving over Isabelle/HOL”. In: *Proc. ACM Program. Lang.* 10.OOPSLA1 (Apr. 2026). DOI: 10.1145/3798275. URL: <https://doi.org/10.1145/3798275>.
- [68] J. Zhang, F. Zhang, J. Lu, J. Hu, X. Yin, L. Zhang, F. Yang, and Y. Zhao. *Towards Real-World Industrial-Scale Verification: LLM-Driven Theorem Proving on seL4*. 2026. URL: <https://arxiv.org/abs/2602.08384>.
- [69] L. Zhang, S. Lu, and N. Duan. “Selene: Pioneering automated proof in software verification”. In: *ACL (Volume 1: Long Papers)*. 2024, pp. 1776–1789.
- [70] L. Zhang, **M. Valentino**, and A. Freitas. “Autoformalization in the Wild: Assessing LLMs on Real-World Mathematical Definitions”. In: *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Ed. by C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng. Suzhou, China: Association for Computational Linguistics, Nov. 2025, pp. 1720–1738. ISBN: 979-8-89176-332-6. DOI: 10.18653/v1/2025.emnlp-main.90.
- [71] Y. Zhao, D. Sanán, F. Zhang, and Y. Liu. “Reasoning about information flow security of separation kernels with channel-based communication”. In: *TACAS*. Springer. 2016, pp. 791–810.

## Appendix: Proofcraft Letter of Support

To Whom It May Concern



Dr Gerwin Klein  
Chief Scientist, Proofcraft Pty Ltd

### Letter of Support for the ARIA proposal Agentic-seL4

Sydney, 26 Jun 2026

With this letter we would like to express our strong support for the ARIA project proposal **“Agentic-seL4: Capability-Based Sandboxing of AI Agents”**.

Not only does this proposal address a clearly important and urgent topic, it is also backed by an excellent team with deep experience in the research questions to be solved.

As the maintainers and initial developers of the seL4 proof base, we are well placed to comment on the suitability of the seL4 microkernel for the isolation requirements for LLM agents described in the proposal, as well as the feasibility and need for improving the seL4 information flow proofs this will require.

The capability model proposed in Agentic-seL4 is a perfect fit for the dynamic nature of LLM-based workloads, which clearly struggle with traditional static access control models. Building this higher-level capability model on existing formalised and proven seL4 capability access control is likely to significantly reduce the formalisation and implementation burden. Implementing this model in a prototype as proposed, will allow the team to evaluate and iterate on the results. Despite its dynamic capability model, the existing information flow proofs for seL4 assume a completely static system setting. This limitation stems from the state of the art of information flow policies and formalisation in 2013 when these seL4 proofs were first developed — pioneered by Prof Toby Murray, who is part of the proposal team. The extension to dynamic information flow policies fits the goals of isolating dynamic LLM agents perfectly.

The seL4 proofs are likely the largest continually maintained mechanised proofs in existence in formal software verification. Like the seL4 kernel itself, the proofs are open source and available for free to academia and industry. The modernisation and potential engineering improvements the proposal aims at have the potential to substantially increase the agility and application of the seL4 proofs, and with these of formal methods industry-wide, because the isolation offered by the kernel enables further, compositional verification to take place on user-level applications.

Overall, Proofcraft and the seL4 Foundation will be following this project with great interest and we are looking forward to the results it will produce.

With best regards,

Dr Gerwin Klein  
Chief Scientist, Proofcraft Pty Ltd  
Chair, Technical Steering Committee of the seL4 Foundation

## Appendix: University of Melbourne Letter of Support



Professor Brijesh Dongol, Project Lead  
University of Surrey  
30 June 2026

### Institutional Letter of Support

#### **Agentic-seL4: Provably Secure Capability-Based Sandboxing of AI Agents**

The University of Melbourne supports Professors Toby Murray and Cezary Kaliszyk participating in the proposal titled **Agentic-seL4: Provably Secure Capability-Based Sandboxing of AI Agents**. The University confirms its commitment to form an agreement in support of this project, contingent upon the agreement of terms to be established between the parties. This statement is not intended to create legal or binding obligations on either party. It is intended as a record of the University's current intentions only.

Ranked #1 in Australia, the University of Melbourne possesses world class research infrastructure, facilities and expertise that is fundamental to the objectives of the project. The University has strategically aligned its research expertise, research environments and support staff to meet our mission of delivering transformative impact in education and research, as outlined in the University's *Advancing Melbourne 2030* strategy. The strategy highlights the institution's commitment to lead, convene and collaborate through strategic partnerships on a global scale.

Professors Murray and Kaliszyk bring unique expertise to the proposed project. Professor Murray led the team who completed the original landmark security proofs for seL4. He also led the development of verifiably secure seL4-based systems for protecting sensitive information. This work was recognised by highly prestigious awards including the 2021 Australian Museum Eureka Prize for Outstanding Science in Safeguarding Australia and the 2022 ACM Software System Award. Professor Kaliszyk brings world-leading expertise on leveraging Artificial Intelligence (AI) for automating formal proof development, including in Isabelle/HOL. His research includes pioneering agentic-based methods for automating interactive proof development, building on many years developing AI-based methods for guiding automated formal reasoning.

The University of Melbourne and all named project personnel understand and accept the roles and terms specified by the sponsor. Should the proposal be successful, the University of Melbourne is committed to providing the applicants with the necessary administrative and other support as detailed in the application to achieve project objectives and meet sponsor requirements.

Yours sincerely,

Laura Arfi  
Manager, Research Contracts  
Office of Research Management  
University of Melbourne